SPPU Engineers — Your vision, Our Innovation

MASTER SOLUTION — UNIT I: INTRODUCTION TO COMPUTER NETWORKS AND PHYSICAL LAYER		
Subject: Computer Networks	Branch: Computer Engineering	
Pattern: 2024	Sem: V	Total Questions: 15
Introduction to computer networks; uses of computer networks – business applications, home applications, mobile users; network hardware – PAN, LAN, MAN, WAN and internetworks; network software – protocol hierarchies, design issues for layers, connection-oriented and connection-less services, service primitives, relationship between services and protocols; reference models – OSI and TCP/IP models. Physical Layer: guided transmission media; wireless transmission; telephone system; narrowband and broadband communication systems		

Q1	**Explain the basic building blocks and utility of computer networks. Describe business, home, and mobile applications of computer networks with suitable examples. [8M]**

Ans. A computer network is an interconnected collection of autonomous computing devices that are capable of exchanging information and sharing resources with one another using a common set of communication rules called protocols. The devices, referred to as nodes or hosts, are connected through transmission links such as copper cable, optical fibre or wireless radio channels, and the overall arrangement allows data, hardware and software resources to be used jointly rather than in isolation.

The fundamental building blocks of a computer network can be understood in terms of the elements that must be present before any communication can take place. These elements work together so that a message generated at a source node reaches the intended destination node correctly and efficiently.

- **End Devices (Nodes/Hosts):** These are the computers, servers, smartphones, printers and other terminal devices that originate or receive data. Every end device is assigned a unique address so that it can be identified uniquely on the network.
- **Transmission Media:** This is the physical or wireless path through which signals travel between devices. It may be guided, such as twisted pair, coaxial cable and optical fibre, or unguided, such as radio waves, microwave and infrared.
- **Networking Devices:** Hubs, switches, routers, bridges and gateways are intermediate devices that forward, filter or translate data as it moves from the source to the destination across one or more networks.
- **Protocols:** A protocol is a formal set of rules that governs the format, timing, sequencing and error control of data exchanged between communicating entities. Without an agreed protocol, two devices cannot interpret each other's signals correctly.

- **Network Software:** Operating system components, network interface drivers and application-layer software together implement the protocol stack and present networking services to the end user in a usable form.

The utility of computer networks arises because they allow geographically distributed users and machines to share resources, communicate instantly and access common data. This sharing improves efficiency, reduces the cost of duplicating expensive resources, and increases the reliability of an organisation because data can be replicated across multiple machines. Based on who uses these benefits and how, network applications are broadly classified into business applications, home applications and mobile user applications.

- **Business Applications:** In a corporate environment, networks primarily target resource sharing and efficient communication among employees. A common example is a client-server model where several client machines access a centralised database server for order processing, payroll or inventory management, avoiding duplication of records at every desk.
- **Business-to-Business Commerce:** Organisations use networks to place purchase orders, verify invoices and track shipments electronically with suppliers, cutting down paperwork and turnaround time. Electronic Data Interchange (EDI) over a network is a classic example used by large manufacturing supply chains.
- **Business-to-Consumer Commerce:** Companies expose their catalogues and services over the network so that customers can browse, order and make payments online, as seen in e-commerce portals such as Amazon and Flipkart.
- **Collaborative Working:** Networks enable video conferencing, shared document editing and groupware applications so that geographically separated teams can work jointly on the same project in real time.
- **Home Applications:** At the household level, networks give individual users access to information, entertainment and communication. Access to remote information such as browsing news portals, digital libraries and government websites is one of the earliest and still dominant home uses.
- **Person-to-Person Communication:** Electronic mail, instant messaging platforms such as WhatsApp, and video calling applications such as Zoom allow family members and friends to stay connected irrespective of physical distance.
- **Interactive Entertainment:** Streaming of movies and music through services such as Netflix and Spotify, along with online multiplayer gaming, forms a major share of home network traffic today.
- **Electronic Commerce for Individuals:** Home users make use of online banking, bill payment and shopping applications, all of which depend on a reliable and secure network connection to a remote server.

- **Mobile User Applications:** Mobile users connect to networks through laptops, tablets and smartphones while moving from one location to another, so the network must support handoff between access points or cell towers without breaking the session.
- **Text and Multimedia Messaging:** Short Message Service, multimedia messaging and social networking applications such as Instagram and Facebook allow mobile users to exchange information instantly using cellular or Wi-Fi data.
- **Location Based Services:** Global Positioning System integrated with mobile networks allows navigation applications such as Google Maps to provide turn-by-turn directions and locate nearby services.
- **Mobile Commerce and Banking:** Digital wallets and Unified Payments Interface based applications let a mobile user complete a financial transaction instantly while travelling, relying entirely on the reliability of the underlying wireless network.

In summary, business, home and mobile applications together demonstrate that the true utility of a computer network is not confined to a single environment. The same underlying building blocks of nodes, media, devices and protocols are reused across all these domains, and only the scale, mobility requirement and traffic pattern change from one application category to another. This is precisely why a single organisation can simultaneously run a business database over its LAN, allow employees to check personal mail from home over the same Internet connection, and support field staff accessing the corporate system from mobile devices, all using the identical set of networking principles.

Q2	**Differentiate between LAN, MAN and WAN with respect to area covered, ownership, speed and technology used. [6M]**

Ans. Networks are commonly classified on the basis of the geographical area they span, because the physical distance directly affects the choice of transmission technology, the achievable data rate and the ownership pattern of the infrastructure. The three principal categories used for this classification are Local Area Network (LAN), Metropolitan Area Network (MAN) and Wide Area Network (WAN).

- **Local Area Network (LAN):** A LAN connects computers within a single building, floor or campus, typically spanning a few hundred metres to a few kilometres. It is owned privately by the organisation that uses it and offers very high data rates with low error rates because the transmission distance is short.
- **Metropolitan Area Network (MAN):** A MAN covers a larger geographical area such as an entire city, typically up to about 50 kilometres. It is often used to interconnect several LANs belonging to different branches of the same organisation and may be owned by a single company or leased from a service provider such as a cable television or telephone company.
- **Wide Area Network (WAN):** A WAN spans a country, continent or even the whole globe, as in the case of the Internet. Because it covers very large distances, it is generally owned

collectively by multiple organisations or leased from telecommunication carriers rather than owned by a single private entity.

Sr. No.	Parameter	LAN	MAN	WAN
1	Geographical Area	Building or campus, up to a few kilometres	Entire city, up to about 50 km	Country, continent or global
2	Ownership	Privately owned by a single organisation	Owned by one organisation or a service provider	Owned collectively or leased from carriers
3	Data Rate	Very high, typically 100 Mbps to several Gbps	Moderate to high, typically tens of Mbps to Gbps	Comparatively lower, though backbone links can be high speed
4	Transmission Technology	Twisted pair, coaxial cable, fibre or Wi-Fi	Fibre optic backbone, cable modem or WiMAX	Leased telephone lines, satellite links, fibre backbone
5	Error Rate	Low, due to short distance and controlled environment	Moderate, since public infrastructure is often used	Comparatively higher because of long distance and multiple hops
6	Design Complexity	Simple, single administrative domain	Moderate, multiple LANs interconnected	Complex, involves routing across many networks and domains
7	Example	Office LAN, college computer laboratory	City cable television network, WiMAX network	Internet, banking network across branches nationwide

From the comparison, it is evident that as the geographical span increases from LAN to WAN, the achievable speed generally decreases while the ownership becomes more distributed and the design complexity increases due to the involvement of multiple intermediate networks and routing decisions. A network designer therefore selects LAN technology for a single office, MAN technology for city-wide connectivity, and WAN technology when nationwide or global reach is required.

It is also worth noting that the boundary between these categories has become somewhat blurred in modern networking, since technologies such as fibre-to-the-home broadband and metro Ethernet allow WAN-like reach with LAN-like speeds. Even so, the classification remains useful because it captures the essential trade-off between coverage area, ownership model and achievable performance that every network designer must consider very carefully before selecting the underlying technology for a new deployment across a growing enterprise with multiple sites and offices.

Q3	**Explain the concept of internetworks. Describe how PAN, LAN, MAN and WAN are interconnected to form larger networks. [6M]**

Ans. An internetwork, often shortened to internet with a lowercase i, is formed when two or more individual networks that may use different underlying technologies are connected together using intermediate devices such as routers or gateways, so that a host on one network can communicate with a host on another network as if they belonged to a single logical network.

The need for internetworking arises because a real-world organisation rarely uses only one type of network. A company may have a Personal Area Network around an executive's desk, a Local Area Network within a department, a Metropolitan Area Network connecting its branches within a city, and a Wide Area Network linking its offices across the country. These distinct networks must cooperate seamlessly, and this cooperation is what internetworking achieves.

- **Personal Area Network (PAN):** A PAN is the smallest network, typically covering a range of a few metres around an individual, connecting devices such as a smartphone, laptop, wireless headset and smartwatch using Bluetooth or infrared. A PAN forms the innermost layer of the internetwork hierarchy.
- **Local Area Network (LAN):** Multiple PANs and standalone hosts within an office or floor are connected to a common LAN through switches. The LAN acts as the first level of aggregation, gathering traffic from many individual devices onto a shared high speed backbone.
- **Metropolitan Area Network (MAN):** Several LANs belonging to different branches of the same organisation within a city are interconnected through a MAN, usually built over a fibre optic or cable backbone maintained by a service provider. Routers at the edge of each LAN forward inter-branch traffic onto the MAN.
- **Wide Area Network (WAN):** MANs in different cities, or LANs directly, are finally interconnected through a WAN that may use leased lines, satellite links or the public Internet backbone. Core routers operated by Internet Service Providers exchange routing information so that a packet can be forwarded across several intermediate networks to reach a distant destination.

The interconnection of PAN, LAN, MAN and WAN into one internetwork is made possible by routers, which operate at the network layer and use routing protocols to decide the next hop for every incoming packet, and by gateways, which additionally perform protocol translation when the interconnected networks use incompatible technologies at the lower layers. Addressing schemes such as IP addressing give every device a globally unique identity so that a packet generated inside a PAN can be correctly delivered to a destination residing inside a distant WAN, regardless of how many intermediate networks it must traverse.

Fig — Hierarchical Internetwork formed by PAN, LAN, MAN and WAN

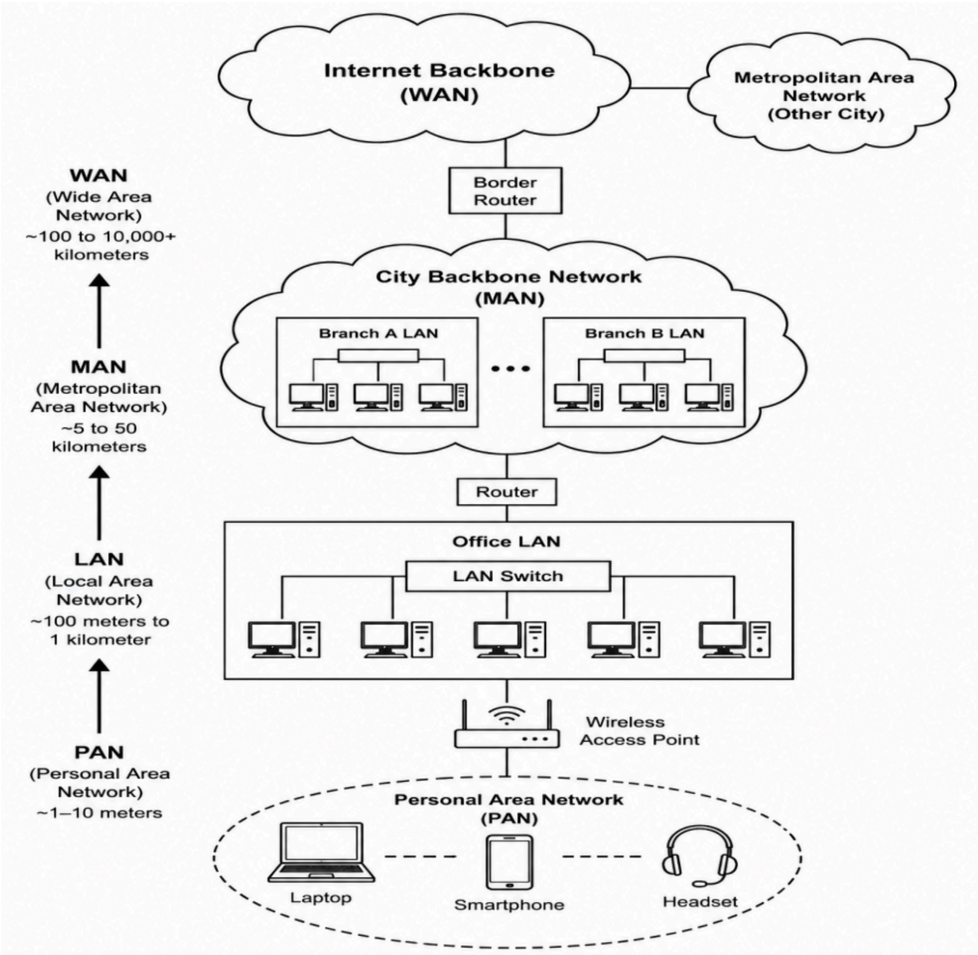


Figure: Hierarchical Relationship of PAN, LAN, MAN and WAN

Q4	**Explain protocol hierarchies and the design issues faced while designing layers of a computer network. [8M]**
Ans.	To reduce the complexity involved in designing a complete network, most networks are organised as a series of layers or levels, each one built upon the layer below it. Every layer offers certain services to the layer directly above it while hiding the internal details of how those services are actually implemented, an idea known as a protocol hierarchy. Within a protocol hierarchy, the set of layers and protocols is together called the network architecture. The rules and conventions used for conversation between two corresponding layers on different machines constitute the protocol for that layer, whereas the boundary between adjacent layers on the same machine defines an interface, which specifies what primitive operations and services the lower layer makes available to the upper layer.

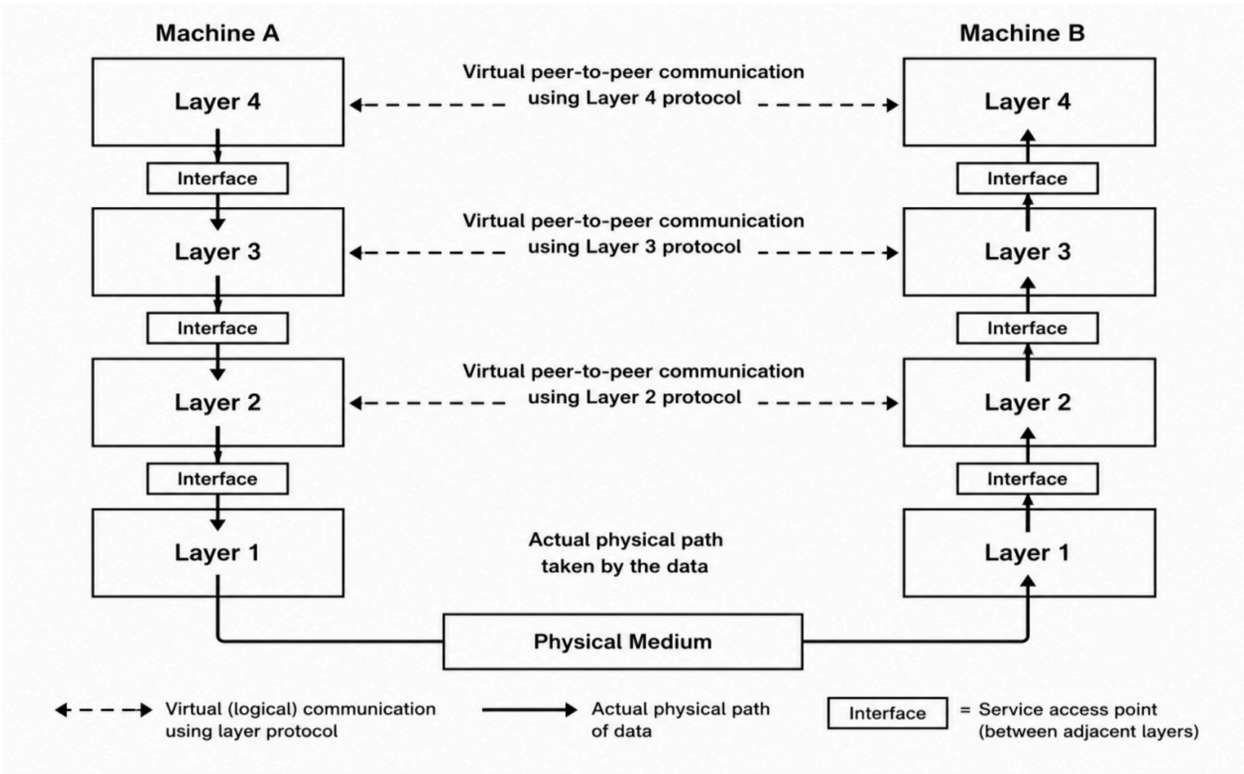
- **Peer-to-Peer Communication:** The entities occupying the same layer on different machines are called peers. Peers communicate using the layer n protocol, but the actual data does not travel directly between them; instead it is passed down through the lower layers, transmitted over the physical medium, and then passed up through the corresponding layers at the receiving machine, an arrangement known as virtual communication supported by actual physical communication at the lowest layer.
- **Encapsulation:** When data moves from an upper layer to a lower layer for transmission, each layer adds its own header, and sometimes a trailer, containing control information required by its peer at the destination. This wrapping of data inside successive headers as it descends the stack is called encapsulation, and the reverse process of removing headers as data ascends the stack at the receiver is called decapsulation.

While the layering concept simplifies design, the actual protocols at each layer must still resolve several fundamental design issues that are common to almost every layer of a network architecture.

- **Addressing:** Every layer needs a mechanism to identify the sender and the intended receiver uniquely, since a system may support multiple conversations at the same time and messages must be delivered to the correct process and machine.
- **Error Control:** Because physical transmission media are inherently imperfect, layers must agree on techniques to detect and, where necessary, correct errors such as bit corruption that occur during transmission, using methods like parity checks, checksums or cyclic redundancy checks.
- **Flow Control:** A fast sender can easily overwhelm a slow receiver if data is transmitted without regard to the receiver's processing or buffering capacity, so a flow control mechanism, either feedback based or rate based, must be designed to prevent this.
- **Sequencing and Reassembly:** When a long message is broken into smaller units for transmission, these units may arrive out of order because different pieces can travel via different routes, so a mechanism to reassemble them correctly at the destination is essential.
- **Multiplexing and Demultiplexing:** A single physical connection is often shared by multiple independent conversations, so the layer must provide a multiplexing scheme to combine several streams for transmission and a demultiplexing scheme to separate them correctly at the receiver.
- **Routing:** When multiple paths exist between a source and a destination, the layer must decide which path to use, and this decision may need to adapt dynamically if part of the network becomes congested or fails.
- **Connection Establishment and Release:** For connection oriented services, layers must decide how to set up a logical connection before data transfer and how to gracefully release it afterwards, ensuring that no data is lost during either phase.
- **Choice between Connection-Oriented and Connectionless Service:** Every layer must decide whether it will offer a connection-oriented service that guarantees ordered and reliable delivery at the cost of extra setup overhead, or a simpler connectionless service that trades

reliability for lower delay and reduced complexity, a decision that strongly affects every layer built above it.

Fig — Protocol Hierarchy showing peer communication and interfaces



Together, protocol hierarchies and their associated design issues provide the theoretical foundation on which practical reference models such as OSI and TCP/IP are built, allowing network designers to develop, test and upgrade any single layer without disturbing the working of every other layer in the stack.

Q5	**Differentiate between connection-oriented and connectionless services. Explain the service primitives used to implement a connection-oriented service. [8M]**

Ans. Every layer in a network architecture can offer two fundamentally different types of service to the layer above it, namely connection-oriented service and connectionless service, and the choice between them affects reliability, ordering and the overhead involved in communication.

A connection-oriented service works on the analogy of the telephone system. Before any actual data can be exchanged, the sender and receiver must first establish a dedicated logical connection, use it to transfer data reliably and in the correct order, and finally release it once communication is complete.

A connectionless service, by contrast, works on the analogy of the postal system. Each message, called a datagram, is transmitted independently and carries the full destination address, so no prior setup is required, but the individual datagrams may arrive out of order, be duplicated, or even be lost without any guarantee from the network layer itself.

Sr. No.	Parameter	Connection-Oriented Service	Connectionless Service
1	Prior Setup	Requires connection establishment before data transfer	No setup required, data sent directly
2	Analogy	Similar to the telephone system	Similar to the postal system
3	Ordering of Data	Data delivered in the same order as sent	Data may arrive out of order
4	Reliability	Generally reliable, with acknowledgements and retransmission	Generally unreliable, best effort delivery
5	Overhead	Higher, due to connection setup and teardown	Lower, since no connection state is maintained
6	Resource Reservation	Path and buffers may be reserved for the connection	No dedicated resources reserved in advance
7	Example Protocol	Transmission Control Protocol (TCP)	User Datagram Protocol (UDP), IP

To make a connection-oriented service usable by the layer above, the service is expressed to the user as a set of primitives, which are operations or system calls that the user process can invoke to request the service. The typical service primitives used to implement a connection-oriented service are as follows.

- **LISTEN:** This primitive is invoked by the server process to indicate that it is ready to accept an incoming connection request. The server blocks until some client attempts to connect, effectively announcing its willingness to communicate.
- **CONNECT:** This primitive is invoked by the client process to actively establish a connection with a specified remote server. It typically triggers a connection request packet to be sent to the server and blocks until the connection is accepted or rejected.
- **RECEIVE:** This primitive is used by a process to indicate that it is ready to accept incoming data. It may block the calling process until data actually arrives from the peer over the established connection.
- **SEND:** This primitive is invoked by a process to transmit data over an already-established connection to its peer, and the underlying protocol takes responsibility for reliable and ordered delivery of this data.
- **DISCONNECT:** This primitive is used by either side to terminate the connection once the data transfer is complete. Depending on the protocol, disconnection may be symmetric, requiring both sides to agree, or asymmetric, allowing either side to close it unilaterally.

These five primitives together illustrate the typical sequence followed in a connection-oriented exchange: the server executes LISTEN and then RECEIVE, while the client executes CONNECT followed by SEND, after which either party can invoke DISCONNECT once the required data has been transferred, exactly as seen in a simple client-server socket-based application built over TCP.

In an actual implementation such as the Berkeley socket interface, these abstract primitives map directly onto system calls. The server process first calls SOCKET to create an endpoint, then BIND to associate it with a known port number, then LISTEN to start waiting for connection requests, and finally ACCEPT, which corresponds to the abstract RECEIVE primitive and blocks until a client establishes contact. The client process calls SOCKET followed by CONNECT, which corresponds to the abstract CONNECT primitive described above and triggers the underlying protocol, typically TCP, to perform a three-way handshake with the server before the connection is considered fully established.

Sr. No.	Primitive	Invoked By	Purpose
1	LISTEN	Server	Announce willingness to accept an incoming connection
2	CONNECT	Client	Actively request a connection to a specified server
3	RECEIVE	Server or Client	Block until data or a connection request arrives
4	SEND	Client or Server	Transmit data reliably over the established connection
5	DISCONNECT	Either Party	Terminate the connection after data transfer is complete

The choice between connection-oriented and connectionless service ultimately depends on the nature of the application. Applications such as file transfer and email require guaranteed, ordered delivery and therefore use a connection-oriented transport protocol, whereas applications such as live video streaming and Domain Name System queries can tolerate occasional loss in exchange for lower delay and therefore rely on a connectionless transport protocol. This flexibility to select the appropriate type of service at the transport layer, independent of the choice made by the network layer beneath it, is itself a direct benefit of the layered design discussed earlier, since each layer remains free to offer whichever type of service best suits the applications and users that depend on it for correct and efficient day-to-day operation.

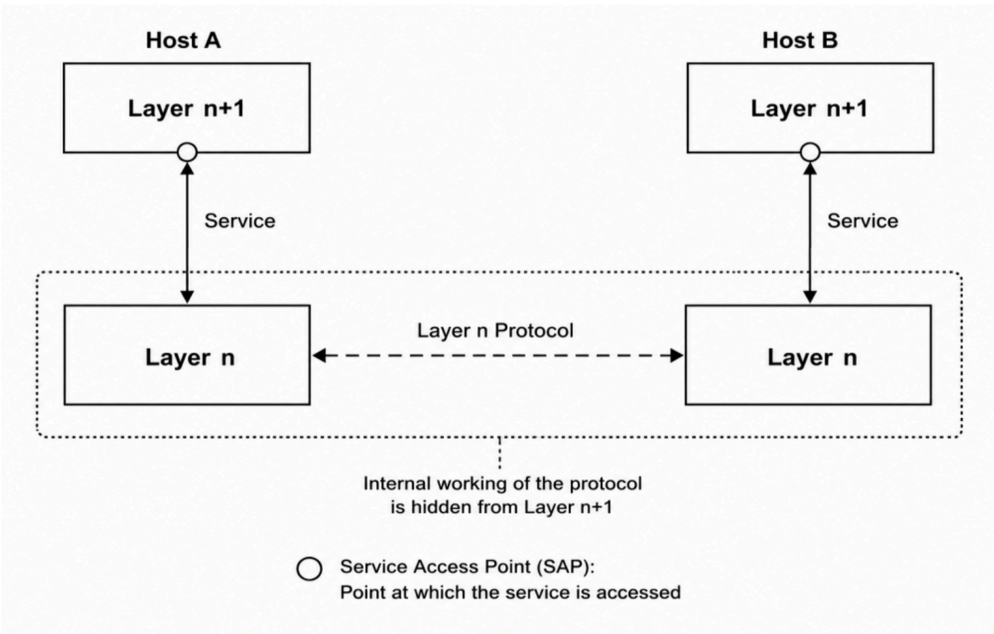
Q6	**Explain the relationship between services and protocols in a layered network architecture with a suitable diagram. [6M]**

Ans. In a layered network architecture, the terms service and protocol are closely related but describe two entirely different aspects of the same layer, and confusing the two is one of the most common conceptual mistakes made while studying network models.

- **Service:** A service is a set of operations that a layer provides to the layer immediately above it. The service defines what the layer can do for its user, but it says nothing about how the operations are implemented internally, so it deals purely with the interface between two adjacent layers on the same machine.
- **Protocol:** A protocol, on the other hand, is a set of rules governing the format and meaning of the frames, packets or messages that are exchanged between peer entities located at the same layer but on different machines. A layer uses its own protocol to implement the service that it advertises to the layer above.

The relationship between the two can be summarised through three important observations. First, services relate to the interface between two layers on the same machine, whereas protocols relate to communication between the same layer on two different machines. Second, a given service can be implemented by more than one protocol, and conversely a protocol can be modified internally as long as the same service continues to be offered upward, meaning that the service and the protocol that implements it can be changed independently of each other. Third, a layer uses the services offered by the layer directly below it in order to implement its own protocol and, consequently, its own service to the layer above.

Fig — Relationship between Service and Protocol across two adjacent layers



This clear separation between service and protocol is what allows network designers to redesign the internal protocol of any one layer, for example replacing an older routing protocol with a more efficient one, without requiring any change to the layers situated above it, as long as the same set of services continues to be exposed at the interface.

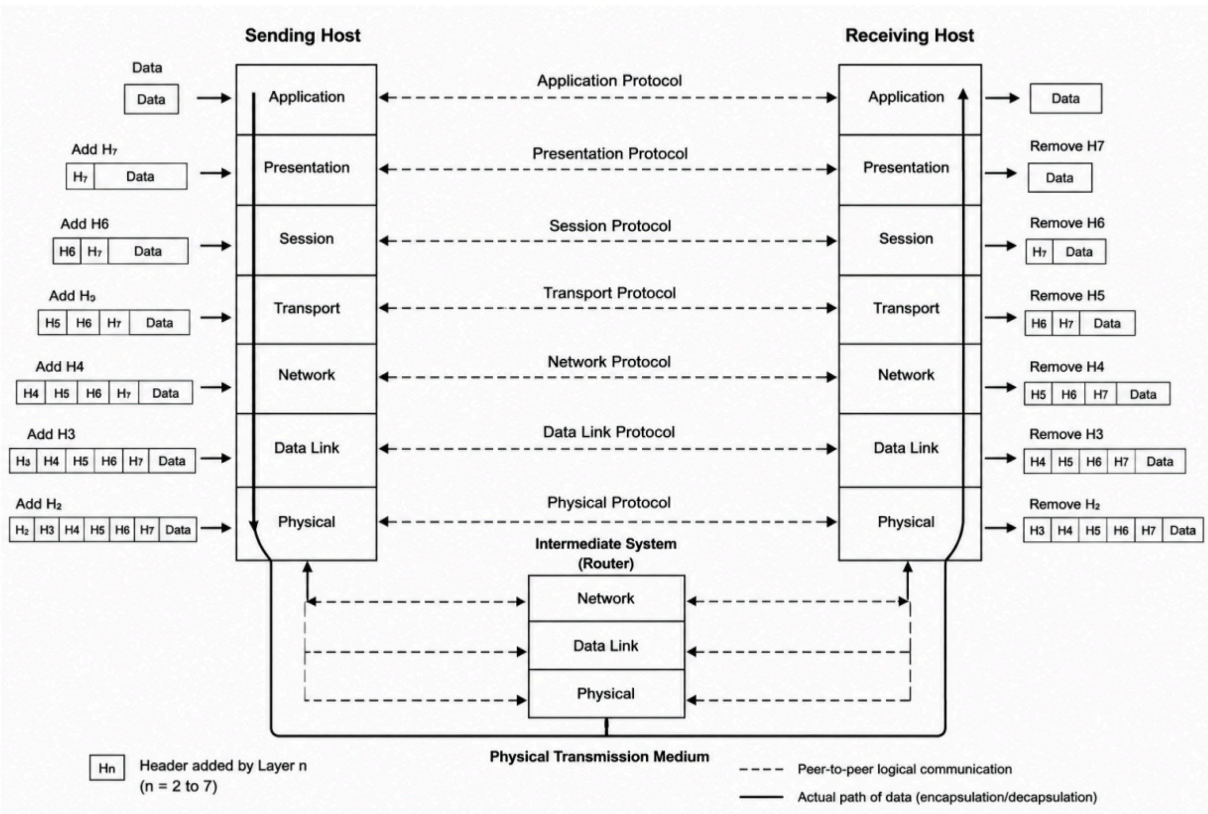
A concrete example makes this distinction clearer. The transport layer offers a reliable byte-stream service to the application layer above it, and this same service can be implemented internally using different transport protocols or even different congestion control algorithms without the application ever being aware of the change, since the application only interacts with the service primitives such as SEND and RECEIVE and never directly with the internal protocol mechanics.

Q7 **Draw and explain the OSI reference model, stating the function of each layer. [10M]**

Ans. The Open Systems Interconnection reference model, commonly called the OSI model, was developed by the International Organization for Standardization as a conceptual framework to standardise the functions of a communication system into seven distinct layers. It was designed so that products from different manufacturers, each implementing the same layer functions, could interoperate with one another.

The OSI model follows the general principle of a layered architecture in which each layer performs a well-defined function, offers a service to the layer above it, and communicates with its peer layer on the remote machine using an agreed protocol. Data generated at the topmost layer of the sender travels down through every layer, gaining a header at each stage, crosses the physical medium, and then travels up through the corresponding layers at the receiver, where each header is removed in reverse order.

Fig — Seven Layer OSI Reference Model



- **Physical Layer:** This is the lowest layer and is responsible for the actual transmission of raw, unstructured bit streams over a physical medium. It defines the electrical, mechanical and timing characteristics such as voltage levels, cable specifications, connector shapes and data rates.
- **Data Link Layer:** This layer takes the raw bit stream from the physical layer and organises it into structured units called frames. It is responsible for framing, physical addressing using MAC addresses, error detection and correction within a single link, and flow control between two directly connected nodes.
- **Network Layer:** This layer is responsible for delivering packets from the source host to the destination host, potentially across multiple intermediate networks. Its main functions are logical addressing using IP addresses, routing of packets along the best available path, and congestion control at the network level.
- **Transport Layer:** This layer provides end-to-end delivery of the complete message between the source and destination processes, taking care of segmentation of data into smaller units, reassembly at the receiver, flow control, error control and, where required, connection-oriented reliable delivery.
- **Session Layer:** This layer establishes, manages and terminates sessions between two communicating application processes. It provides services such as dialogue control, deciding whether communication is half duplex or full duplex, and synchronisation using checkpoints so that a long transfer can resume from the last checkpoint after a failure.
- **Presentation Layer:** This layer is concerned with the syntax and semantics of the information exchanged, handling tasks such as data translation between different representation formats, data compression to reduce the number of bits transmitted, and encryption to protect data from unauthorised access.
- **Application Layer:** This is the topmost layer and provides the interface directly to the end user, supporting services such as file transfer, electronic mail, web browsing and remote login through protocols like HTTP, FTP, SMTP and DNS.

A commonly used memory aid for remembering the seven layers from the physical layer upward is the sentence Please Do Not Throw Sausage Pizza Away, corresponding to Physical, Data Link, Network, Transport, Session, Presentation and Application. Although the OSI model is now used mainly as a teaching and reference tool rather than as a directly implemented protocol suite, it remains extremely important because it provides a precise vocabulary for discussing network functions and is the standard against which practical architectures such as TCP/IP are usually compared.

Sr. No.	Layer	Protocol Data Unit	Typical Device Operating at this Layer
1	Application	Message	End-user application software

2	Presentation	Message	End-user application software
3	Session	Message	End-user application software
4	Transport	Segment	Firewall performing deep packet inspection
5	Network	Packet	Router
6	Data Link	Frame	Switch, Bridge
7	Physical	Bit	Hub, Repeater, Cable

Each layer of the OSI model also defines its own protocol data unit, which is simply the name given to the block of information handed down or up between adjacent layers. Knowing the name of the protocol data unit at every layer, together with the typical networking device that operates at that layer, is extremely useful in practice because it tells a network engineer exactly which device needs to be examined when troubleshooting a fault; for instance, a cabling fault is diagnosed at the physical layer using a cable tester, whereas a routing loop is diagnosed at the network layer using a router's routing table.

One important critique of the OSI model, despite its conceptual elegance, is that it was published only after the protocols had already been designed to fit the model rather than the reverse, and consequently very few real networks today implement all seven layers exactly as separately defined; in practice, the session and presentation layer functions are frequently folded into the application itself, which is exactly the simplification adopted by the more pragmatic four-layer TCP/IP model discussed subsequently.

Despite this practical limitation, the seven-layer breakdown remains extremely valuable for structured fault diagnosis in the field. A network administrator troubleshooting a connectivity complaint typically works upward through the stack in order, first verifying that the physical cabling and link lights are functioning correctly, then confirming that switches are forwarding frames correctly at the data link layer, then checking that routers hold correct routing table entries at the network layer, and only then examining transport-layer port availability and application-layer configuration, a disciplined bottom-up approach that is a direct and lasting benefit of thinking about a network in terms of the OSI layers.

The OSI model also standardises important terminology that is used across the networking industry regardless of which actual protocol suite is deployed, such as describing a switch as a layer 2 device and a router as a layer 3 device, terminology that persists in vendor documentation, certification examinations and product datasheets even though the underlying protocols in use today are almost always drawn from the TCP/IP suite rather than from the original OSI protocol stack. This lasting influence is why every serious study of computer networking, including preparation for university examinations and professional networking certifications, still begins with a thorough understanding

	of the seven-layer OSI reference model before moving on properly to the practical, widely deployed TCP/IP protocol suite that actually runs the modern Internet on a day-to-day basis, worldwide.

Q8	Explain the TCP/IP reference model and compare it with the OSI reference model. [8M]

Ans. The TCP/IP reference model was developed by the United States Department of Defense to interconnect multiple heterogeneous networks reliably, even in the presence of partial failures such as the loss of a subnet during wartime conditions. Unlike the OSI model, which was designed first as a theoretical framework, the TCP/IP model grew out of the actual protocols that were already in use on the ARPANET and the early Internet.

- **Host-to-Network Layer:** This is the lowest layer of the TCP/IP model and corresponds broadly to the physical and data link layers of OSI. It is responsible for the actual transmission of bits over the underlying network hardware, but the TCP/IP model itself does not specify this layer in detail, leaving it to be defined by the specific network technology used, such as Ethernet or Wi-Fi.
- **Internet Layer:** This layer is the backbone of the TCP/IP model and is responsible for allowing hosts to inject packets into any network and have them travel independently to the destination, possibly via different networks. Its core protocol is the Internet Protocol, which handles logical addressing and routing, along with supporting protocols such as ICMP for error reporting.
- **Transport Layer:** This layer is designed to allow peer entities on the source and destination hosts to carry on a conversation, exactly as in the OSI transport layer. It defines two main end-to-end protocols, the connection-oriented Transmission Control Protocol which provides reliable, ordered delivery, and the connectionless User Datagram Protocol which provides a fast but unreliable service.
- **Application Layer:** This topmost layer combines the functions of the OSI session, presentation and application layers into a single layer, since the designers of TCP/IP felt that these functions were rarely used separately in practice. It hosts high level protocols such as HTTP, FTP, SMTP, DNS and TELNET that directly serve user applications.

Sr. No.	Parameter	OSI Model	TCP/IP Model
1	Number of Layers	Seven layers	Four layers
2	Development Approach	Theoretical model developed first, protocols fitted later	Protocols developed first, model described afterward
3	Session and Presentation Layers	Provided as two separate distinct layers	Not present separately, merged into the application layer

4	Guarantee of Delivery	Guarantees reliable delivery at the network layer	Network layer offers only best-effort connectionless delivery
5	Protocol Dependence	Model is protocol independent, generic in nature	Model is closely tied to specific protocols such as TCP and IP
6	Usage Today	Used mainly as a reference and teaching model	Used as the actual working model of the Internet
7	Layer Coupling	Strict boundary and service definition between every layer	Layers are less strictly separated, with more integration

Overall, both models describe communication using a layered approach and share the same general goal of allowing interoperable communication, but the OSI model is valued for its clean theoretical separation of concerns into seven layers, whereas the TCP/IP model is valued because it describes the protocols that are actually deployed and running the real Internet today. In practice, network engineers frequently use OSI terminology, such as referring to layer 2 switches or layer 3 routing, while relying on the TCP/IP protocol suite for actual data transmission.

An important additional difference lies in how each model treats reliability as a design responsibility. The OSI model builds guaranteed, error-free delivery into the network layer itself, which was a reasonable assumption for the relatively well-controlled telecommunication networks of the era in which OSI was designed. The TCP/IP model, in contrast, was built for the ARPANET, an environment where individual links or nodes could fail unpredictably, so its designers deliberately kept the internet layer connectionless and best-effort, and pushed the responsibility for reliability upward to the transport layer, where the Transmission Control Protocol alone provides retransmission, sequencing and acknowledgement.

Because of this practical origin, most modern textbooks and working network engineers describe real systems using a hybrid five-layer model that keeps the OSI physical and data link layers as separate layers while adopting the TCP/IP internet, transport and application layers, which combines the descriptive clarity of OSI's lower layers with the practical relevance of the TCP/IP upper layers.

Another practical consequence of merging the upper three OSI layers into a single TCP/IP application layer is that functions such as data representation, encryption and session management, which OSI treats as separate standardised layers, are instead implemented individually within each application protocol as needed; for example, the Hypertext Transfer Protocol Secure variant embeds its own encryption directly rather than relying on a separate universal presentation-layer encryption service, illustrating how the TCP/IP model favours pragmatic, protocol-specific solutions over the strict universal layering insisted upon by OSI. This pragmatism is widely regarded as the principal reason the TCP/IP model succeeded commercially, since it allowed the Internet to grow rapidly by letting individual protocols evolve independently to meet immediate practical needs, rather than waiting for a complete and rigorously standardised seven-layer suite to be finalised, ratified and universally adopted by every competing vendor across the industry. This is one of the key reasons the four-layer model is still taught alongside OSI in every modern networking course.

Q9	Explain various guided transmission media used in computer networks. [6M]

Ans. Guided transmission media, also called wired or bound media, provide a physical conducting path along which signals are directed from the sender to the receiver. The choice of guided medium affects the bandwidth, attenuation, susceptibility to interference and cost of a network link.

- **Twisted Pair Cable:** This medium consists of two insulated copper conductors twisted together in a helical shape, and the twisting reduces electromagnetic interference from external sources and crosstalk from adjacent pairs. It is inexpensive and widely used for telephone lines and Ethernet LAN connections, available as unshielded twisted pair and shielded twisted pair variants.
- **Coaxial Cable:** This medium consists of a central copper conductor surrounded by an insulating layer, a metallic shield and an outer protective jacket. Its shielded construction gives it better noise immunity and higher bandwidth than twisted pair, making it suitable for cable television distribution and older Ethernet installations.
- **Fibre Optic Cable:** This medium transmits data as pulses of light through a thin strand of glass or plastic, using the principle of total internal reflection to keep the light confined within the core. It offers extremely high bandwidth, very low attenuation over long distances and complete immunity to electromagnetic interference, making it the medium of choice for backbone and long-haul links.

Sr. No.	Parameter	Twisted Pair	Coaxial Cable	Fibre Optic Cable
1	Bandwidth	Low to moderate	Moderate to high	Very high, in the range of gigahertz
2	Attenuation	High over long distances	Moderate over distance	Very low, suitable for long-haul links
3	Noise Immunity	Susceptible to electromagnetic interference	Better than twisted pair due to shielding	Completely immune to electrical interference
4	Installation Cost	Low cost and easy installation	Moderate cost, bulkier than twisted pair	High cost, requires skilled installation
5	Typical Use	Telephone lines, LAN cabling	Cable television, legacy Ethernet	Internet backbone, long distance trunk lines

Selecting an appropriate guided medium therefore involves a trade-off between cost, required bandwidth, distance to be covered and the level of electrical noise present in the environment, with twisted pair generally preferred for short inexpensive LAN links, coaxial cable for moderate distance broadband distribution, and fibre optic cable wherever very high speed or very long distance transmission is essential.

	Within twisted pair cabling itself, cables are further graded into categories such as Category 3, Category 5e and Category 6, where a higher category number indicates tighter twisting and better manufacturing quality, which allows a higher supported data rate; for example Category 5e cable comfortably supports Gigabit Ethernet over the standard 100 metre segment length used in structured LAN cabling. Fibre optic cable is similarly divided into multimode fibre, which uses a relatively large core and is suited to shorter campus distances, and single mode fibre, which uses a much narrower core and a laser source to achieve extremely long, low-loss transmission distances used in telecom backbone links. Coaxial cable is likewise available in different grades, distinguished mainly by conductor thickness and shielding quality, with thicker cable historically used for long backbone runs and thinner, more flexible cable used for shorter connections closer to the end device. Connector types also differ across these media, with twisted pair typically terminated using RJ-45 connectors, coaxial cable using BNC or F-type connectors, and fibre optic cable using specialised connectors such as SC or LC that must maintain precise optical alignment to avoid excessive signal loss at every joint.

Q10	**Explain wireless transmission and the electromagnetic spectrum used for wireless communication. [6M]**
Ans.	Wireless transmission, also called unguided media, refers to the propagation of electromagnetic signals through free space, air or vacuum without the use of any physical conductor. Since the signal is not confined to a fixed physical path, it can spread in multiple directions and be received by any suitably tuned antenna within range, which makes wireless transmission particularly convenient for mobile users and for locations where laying cable is impractical. All wireless transmission relies on electromagnetic waves that are characterised by their frequency, and different portions of the electromagnetic spectrum exhibit very different propagation behaviour, which is why particular bands are reserved for particular kinds of communication. • **Radio Waves:** Radio waves occupy the lower frequency portion of the spectrum and are easy to generate, can travel long distances and can penetrate buildings, which makes them widely used for indoor Wi-Fi networks, AM and FM broadcasting and cellular mobile communication. • **Microwaves:** Microwaves occupy a higher frequency band than radio waves and travel in a nearly straight line, so they require line-of-sight communication between the transmitting and receiving antennas. They are extensively used for terrestrial point-to-point links and for satellite communication systems that relay signals across continents. • **Infrared Waves:** Infrared waves have an even higher frequency and cannot penetrate solid walls, which actually prevents interference between systems in adjacent rooms. They are used for short range communication such as television remote controls and older laptop-to-printer connections.

Wireless communication systems are typically categorised by their coverage area and mobility support. A wireless personal area network, based on technologies such as Bluetooth, connects devices over a range of a few metres around a single user. A wireless local area network, based on the IEEE 802.11 family of standards commonly known as Wi-Fi, covers a building or campus and allows several devices to share a common access point. A wireless wide area network, implemented through cellular technology, provides coverage over an entire city or region and supports continuous connectivity even as the user moves between different cell towers.

Because wireless transmission is broadcast in nature and travels through a shared, uncontrolled medium, it is more vulnerable to interference, eavesdropping and multipath fading compared to guided media, which is why wireless systems place a much greater emphasis on error control, security mechanisms and medium access control protocols that regulate how multiple devices share the same frequency band without excessive collision.

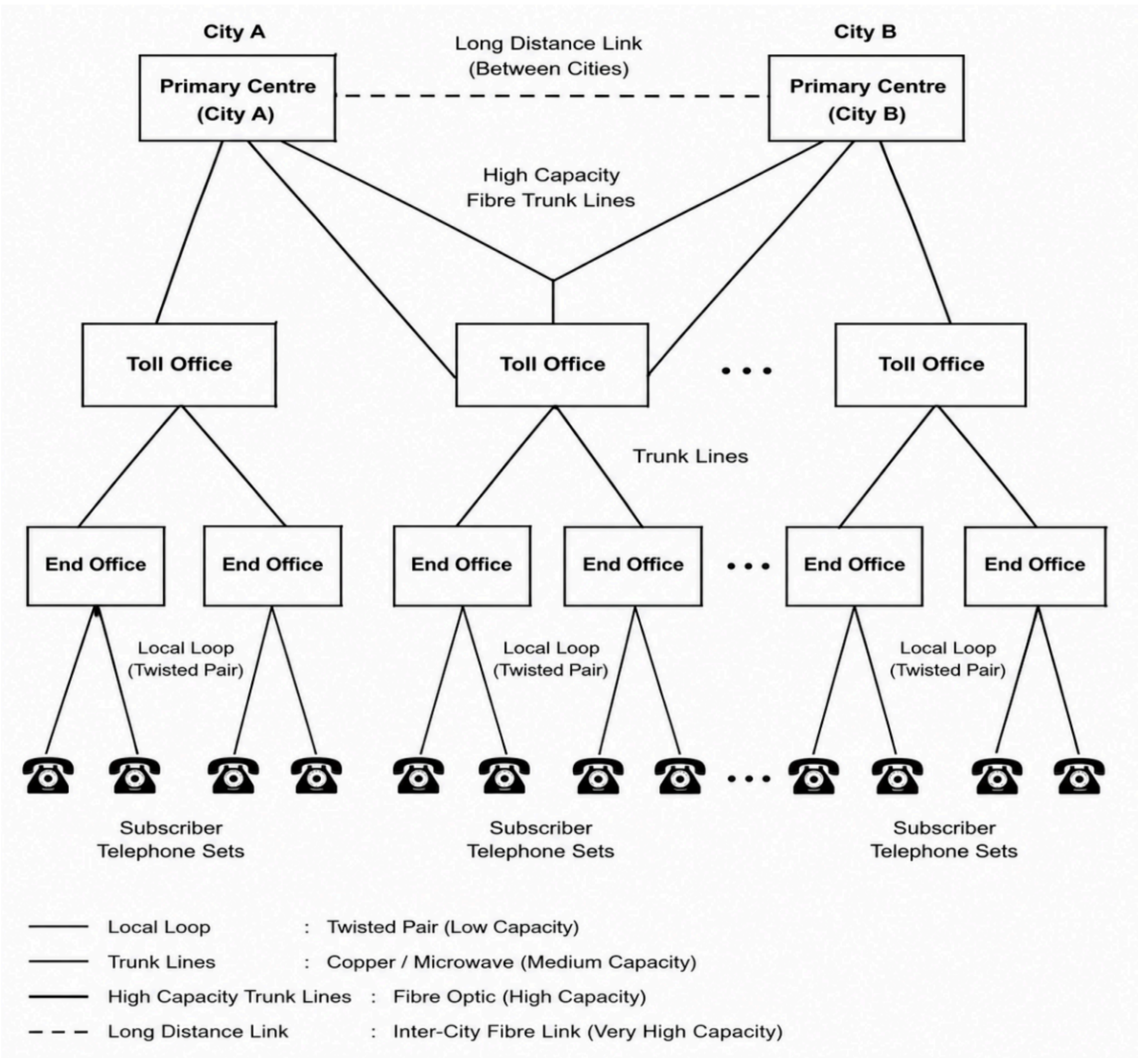
Regulatory bodies in every country allocate specific portions of the electromagnetic spectrum for specific licensed and unlicensed uses to avoid mutual interference between unrelated systems; for example the industrial, scientific and medical band around 2.4 gigahertz is kept unlicensed worldwide and is used by Wi-Fi and Bluetooth devices, whereas the frequency bands used by cellular mobile operators are licensed exclusively to individual telecom companies within a given country, and satellite communication typically uses the higher C-band and Ku-band microwave frequencies to minimise rain-induced signal attenuation. This careful division of the spectrum, overseen internationally by bodies such as the International Telecommunication Union, is essential precisely because wireless transmission is inherently a shared broadcast medium in which uncoordinated use of the same frequency by two independent systems in the same geographic area would otherwise cause severe, unpredictable and hard-to-diagnose mutual interference between unrelated wireless systems operating nearby.

Q11	**Explain the working of the telephone system as a communication network with a suitable diagram. [6M]**
Ans.	The telephone system, originally built to carry analog voice signals, forms one of the oldest and most extensive communication networks in the world, and its infrastructure continues to be reused today to carry digital data, including dial-up and broadband Internet access. At the lowest level, every subscriber's telephone instrument is connected by a pair of copper wires, called the local loop, to the nearest telephone exchange, which is also known as an end office or a central office. When a subscriber picks up the receiver, the exchange detects a change in the electrical loop and provides a dial tone to indicate that it is ready to accept a dialled number. • **Local Loop:** This is the last-mile connection between the subscriber's telephone set and the nearest end office, traditionally made of twisted pair copper cable and is often the limiting factor for the achievable data rate when the same line is later used for broadband access.

- **Trunk Lines:** These are high capacity links, today mostly built on fibre optic cable, that connect one telephone exchange to another. Trunk lines carry the multiplexed traffic of thousands of individual calls simultaneously between exchanges.
- **Switching Offices:** When two subscribers connected to different end offices wish to communicate, the call must pass through one or more intermediate switching offices, which set up a dedicated path between the two end offices for the duration of the call, using circuit switching.

The overall telephone network is organised hierarchically. At the bottom, end offices connect directly to subscriber telephones. A number of end offices are connected to a toll office, several toll offices are connected to a primary centre, and so on, up to the highest level, forming a hierarchy that ensures any subscriber can eventually be connected to any other subscriber in the world through an appropriate sequence of switching offices.

Fig — Hierarchical structure of the Telephone Network



When two subscribers are connected through a call, the switching offices establish a temporary dedicated circuit for the entire duration of the conversation, and this circuit is released only when either party hangs up, which is the defining feature of a circuit switched network as opposed to the packet switched approach used by the modern Internet.

Although the telephone system was originally designed purely for analog voice, its widespread local loop wiring has since been reused to deliver digital data services. Dial-up modems modulate digital data onto an analog voice-band carrier to send it over the same local loop, while Digital Subscriber Line technology uses frequencies above the voice band on the very same copper pair to deliver much higher broadband data rates without disturbing the ordinary telephone conversation, illustrating how the legacy telephone infrastructure continues to underpin modern broadband access.

Q12 **Differentiate between narrowband and broadband communication systems. [5M]**

Ans. Communication systems can be classified according to the width of the frequency band they occupy and the number of simultaneous channels or signals they can support, and this classification broadly divides systems into narrowband and broadband categories.

- **Narrowband Communication:** A narrowband system uses a comparatively small range of frequencies, which is generally sufficient only to carry a single channel of information at a time, such as one voice conversation or one low speed data stream, and the achievable data rate is correspondingly limited.
- **Broadband Communication:** A broadband system uses a much wider range of frequencies and can therefore support multiple channels simultaneously, either by dividing the available bandwidth among several signals or by using techniques that pack far more bits per second onto the same physical medium, resulting in a significantly higher aggregate data rate.

Sr. No.	Parameter	Narrowband	Broadband
1	Bandwidth Used	Small, limited frequency range	Large, wide frequency range
2	Number of Channels	Typically supports a single channel	Supports multiple channels simultaneously
3	Data Rate	Low data rate	High data rate
4	Typical Applications	Traditional telephone voice calls, low speed telemetry	Internet broadband access, cable television, high speed data links
5	Example Technology	Traditional dial-up modem over telephone line	DSL, cable modem, fibre-to-the-home connections

The distinction is important while designing physical layer systems because a narrowband link is adequate for simple low rate applications such as basic telemetry or legacy voice telephony, whereas

	modern applications such as video streaming, cloud computing and large file transfer demand the far higher capacity offered only by broadband communication systems, which is why almost all current Internet access technologies are broadband in nature. It is useful to note that broadband systems achieve their higher aggregate capacity either by dividing the total available bandwidth into several separate frequency sub-channels through frequency division multiplexing, allowing voice, data and video to be carried simultaneously on one physical medium, or by using advanced modulation schemes that pack a very large number of bits into every symbol transmitted over the wide band. Cable television networks are a good illustration of this idea, since a single coaxial or fibre cable simultaneously carries dozens of television channels together with a broadband Internet data channel, all separated from one another purely by frequency. A more recent example illustrating that narrowband is not obsolete is Narrowband Internet of Things, a cellular technology deliberately designed with a very small bandwidth footprint to achieve extremely low power consumption and long range for battery-operated sensor devices, demonstrating that the choice between narrowband and broadband ultimately depends on whether the given application prioritises raw data rate or efficient, economical use of scarce spectrum resources and limited battery power available on the device.

Q13	**Draw Manchester and Differential Manchester encoding for the bit sequence 10110010. Explain the encoding rules used. [6M]**
Ans.	Line coding schemes convert a stream of binary data into an electrical signal suitable for transmission over a physical medium, and Manchester encoding along with Differential Manchester encoding are two widely used self-clocking schemes that ensure a guaranteed voltage transition within every bit period, which allows the receiver to recover the clock directly from the received signal without needing a separate clock line. • **Manchester Encoding Rule:** In standard Manchester encoding, every bit period is divided into two equal halves, and the direction of the transition occurring exactly in the middle of the bit period represents the bit value. A binary 1 is represented by a transition from low to high in the middle of the bit interval, while a binary 0 is represented by a transition from high to low in the middle of the bit interval. • **Differential Manchester Encoding Rule:** In Differential Manchester encoding, there is always a transition in the middle of every bit period purely to aid clock recovery, but the actual bit value is determined instead by the presence or absence of a transition at the very start of the bit period. A binary 0 is represented by a transition at the beginning of the bit interval, whereas a binary 1 is represented by no transition at the start of the bit interval.

Fig — Manchester and Differential Manchester waveforms for 10110010

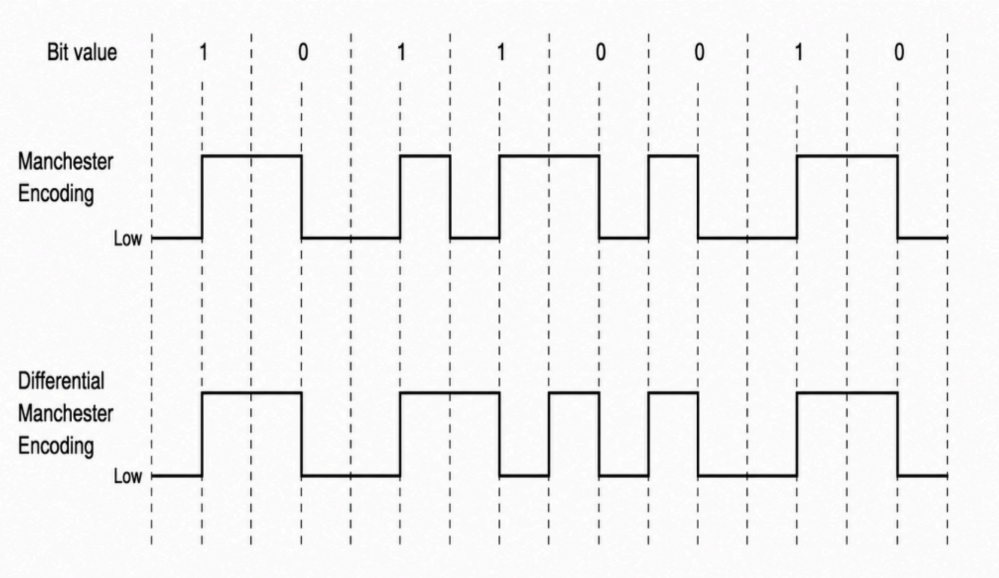


Figure: Manchester and Differential Manchester Encoding

Tracing the given bit sequence 1 0 1 1 0 0 1 0 through Manchester encoding, the waveform shows a low-to-high transition at the centre of the first interval for bit 1, a high-to-low transition at the centre of the second interval for bit 0, low-to-high transitions at the centre of the third and fourth intervals for the two consecutive 1s, high-to-low transitions at the centre of the fifth and sixth intervals for the two consecutive 0s, a low-to-high transition at the centre of the seventh interval for bit 1, and finally a high-to-low transition at the centre of the eighth interval for bit 0.

Tracing the same sequence through Differential Manchester encoding, starting from an assumed low level, every interval shows the mandatory mid-interval transition, and additionally the intervals corresponding to bit 0, namely the second, fifth, sixth and eighth intervals in the given sequence, each begin with an extra transition at the start of the interval, while the intervals corresponding to bit 1, namely the first, third, fourth and seventh intervals, begin with no transition and simply continue from the level left by the previous interval.

Both schemes guarantee at least one transition per bit period, which prevents the long runs of a constant voltage level that would otherwise cause clock drift, but Differential Manchester encoding has the added practical advantage of being insensitive to the polarity of the wire connection, since it depends only on whether a transition is present or absent, and not on the absolute direction of that transition.

Q14	Explain Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) techniques used in wireless transmission. [8M]

Ans. Spread spectrum is a physical layer transmission technique in which the original narrowband signal is deliberately spread across a much wider band of frequency than would normally be required, in order to gain resistance against jamming, interception and narrowband interference. Two of the most widely used spread spectrum techniques are Direct Sequence Spread Spectrum and Frequency Hopping Spread Spectrum.

- **Direct Sequence Spread Spectrum (DSSS):** In DSSS, each data bit of the original signal is combined, using an exclusive-OR operation, with a much faster pseudo-random bit pattern called a chipping code or spreading code, so that every single data bit is represented by multiple chips. Since the chipping rate is far higher than the original data rate, the resulting signal occupies a much wider bandwidth than the original narrowband signal.
- **DSSS Receiver Operation:** At the receiver, the same chipping code that was used at the transmitter is applied again to despread the received signal, which correlates strongly with the intended chipping sequence and recovers the original narrowband data, while any narrowband interference that does not match the chipping code gets spread out and appears as low level background noise after despreading.
- **Frequency Hopping Spread Spectrum (FHSS):** In FHSS, the available frequency band is divided into a large number of smaller sub-channels, and the transmitter and receiver, using a common pseudo-random sequence, hop from one sub-channel to another at fixed, very short time intervals, transmitting only a small burst of data on each sub-channel before hopping again.
- **FHSS Synchronisation:** For successful communication, the receiver must hop through exactly the same sequence of frequencies at exactly the same time as the transmitter, a requirement known as synchronisation, which is typically achieved because both sides share the same pseudo-random number generator seed.

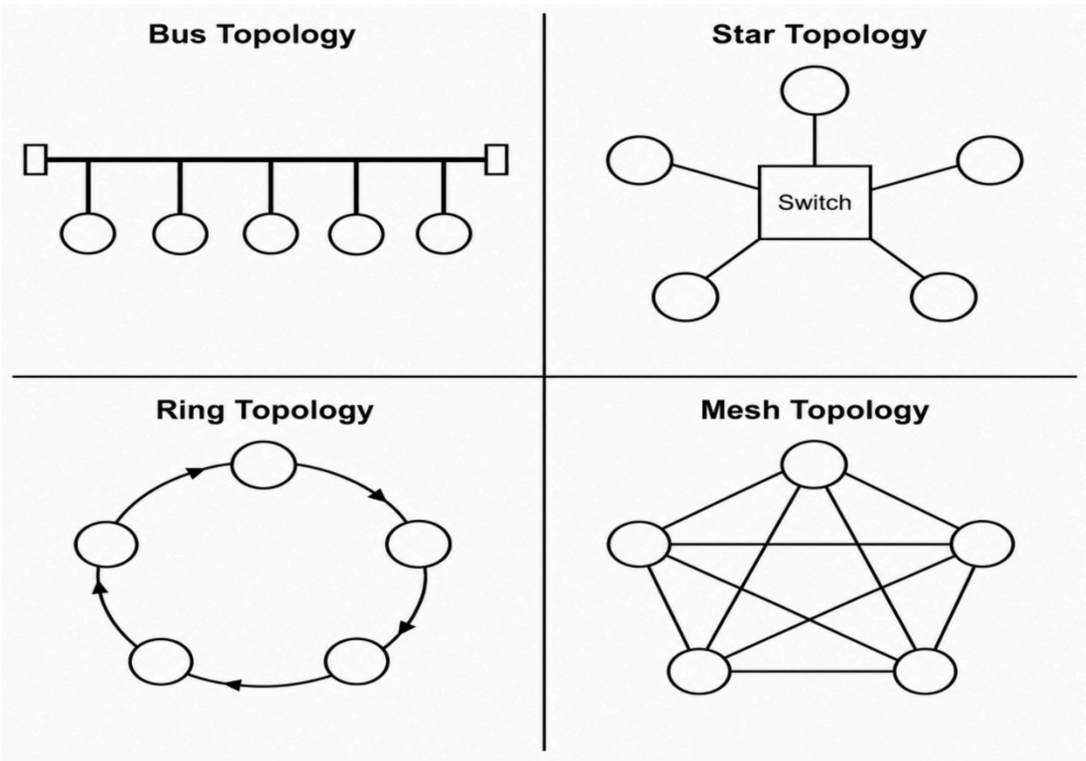
Sr. No.	Parameter	DSSS	FHSS
1	Spreading Method	Multiplies data with a fast chipping code	Hops rapidly across many frequency sub-channels
2	Bandwidth Usage	Uses the entire wide band continuously	Uses only one narrow sub-channel at any instant
3	Resistance to Interference	Resistant because interference is spread out on despreading	Resistant because interference affects only a few hop channels
4	Complexity of Hardware	Requires precise chip-level synchronisation circuitry	Requires a frequency synthesiser to hop quickly
5	Typical Use	Used in older IEEE 802.11b Wi-Fi and CDMA cellular systems	Used in Bluetooth and some early wireless LAN systems
6	Multiple Access Support	Supports code division multiple access using distinct codes	Supports multiple users through distinct hopping patterns

	Both DSSS and FHSS achieve the same underlying goal of spreading the transmitted energy over a wide band so that the signal becomes more difficult to detect, jam or intercept, and both provide graceful degradation in the presence of narrowband interference, but they differ fundamentally in mechanism, with DSSS spreading the signal in the frequency domain simultaneously across the whole band using a code, while FHSS spreads the signal in the time domain by rapidly switching among discrete narrow channels. The origin of spread spectrum technology lies in military communication, where the primary requirement was to make a transmission resistant to deliberate jamming and difficult for an adversary to intercept or locate, since the transmitted energy for any given instant is spread so thinly across the band that it appears merely as background noise to a receiver that does not know the chipping code or hopping pattern being used. This same anti-jamming and low-probability-of-intercept property later made spread spectrum attractive for civilian unlicensed wireless communication, since multiple independent systems using different codes or hopping patterns can coexist in the same frequency band with only a limited increase in mutual interference. In DSSS specifically, the ratio between the chipping rate and the original data rate is called the spreading factor or processing gain, and a higher processing gain gives the system stronger resistance to interference at the cost of consuming more bandwidth for the same original data rate. In FHSS, hopping can be classified as slow frequency hopping, where several data bits are transmitted per hop, or fast frequency hopping, where the system hops multiple times within the transmission of a single data bit, with fast hopping offering greater resistance to jamming at the cost of additional synchronisation overhead. Bluetooth is a widely cited practical example of FHSS, since it hops across seventy-nine channels in the 2.4 gigahertz band up to sixteen hundred times per second to avoid sustained interference with co-located Wi-Fi devices. From an implementation cost perspective, FHSS transceivers are generally simpler and cheaper to build because they only need a fast frequency synthesiser rather than the precise chip-level correlation circuitry demanded by DSSS, which is one reason early low-cost wireless systems such as the original Bluetooth specification favoured frequency hopping, whereas DSSS, despite requiring more complex receiver hardware, delivers a higher effective data rate for a given bandwidth and was therefore preferred in early high-throughput wireless LAN standards and in code division multiple access based cellular systems where many independent users must share the same spectrum simultaneously using their own distinct spreading codes without excessive mutual interference.

Q15	**Explain network topologies such as bus, star, ring and mesh with neat diagrams, along with their advantages and disadvantages. [8M]**
Ans.	Network topology refers to the physical or logical arrangement in which the various nodes of a network are interconnected to one another, and the choice of the physical topology significantly affects the cost, reliability, ease of installation and fault tolerance of the resulting network.

- **Bus Topology:** In a bus topology, all the devices are attached to a single common communication line, called the backbone, through short connector cables called drop lines, and every device receives every transmission but accepts only the frame addressed to it, discarding the rest.
- **Star Topology:** In a star topology, every device is connected individually by a dedicated point-to-point link to a central device, usually a switch or a hub, and all communication between any two devices must pass through this central device.
- **Ring Topology:** In a ring topology, every device is connected to exactly two neighbouring devices, forming a closed loop, and data travels around the ring in one direction, being regenerated and passed along by each device until it reaches the intended destination.
- **Mesh Topology:** In a mesh topology, every device is connected to every other device in the network through a dedicated point-to-point link, so that a message can travel directly from the source to the destination without depending on any intermediate device.

Fig — Bus, Star, Ring and Mesh Topology layouts



Sr. No.	Parameter	Bus	Star	Ring	Mesh
1	Cabling Required	Least cabling, single backbone	Moderate, one link per device to hub	Moderate, one link per neighbour pair	Very high, n(n-1)/2 links required

2	Fault Tolerance	Poor, backbone failure halts entire network	Good, only the failed device is affected	Poor, break in ring can disrupt communication unless dual ring used	Excellent, alternate paths always available
3	Dependency on Central Device	No central device present	Fully dependent on the central hub or switch	No central device, but dependent on ring continuity	No dependency on any single device
4	Ease of Adding a Node	Easy, simply tap into the backbone	Easy, connect new device to a free port on hub	Difficult, ring must be temporarily broken	Very difficult and expensive due to full interconnection
5	Cost	Low cost	Moderate cost, depends on switch capacity	Moderate cost	Very high cost for large networks
6	Typical Use	Small legacy Ethernet LANs	Modern Ethernet LANs using switches	Token ring networks, FDDI backbones	Backbone of the Internet, critical military or telecom links

In practice, the star topology has become the dominant choice for modern local area networks because switch technology has become widely available and inexpensive and the topology localises faults to individual links, whereas mesh topology, despite its excellent fault tolerance, is generally reserved for highly critical backbone connections where the very high cabling cost can be justified by the strong need for guaranteed alternate paths.

Real installations frequently do not restrict themselves to a single pure topology but instead combine two or more basic topologies to obtain the advantages of each, an arrangement generally called a hybrid topology. A typical campus network, for instance, uses a star topology within every individual building, since this localises faults and simplifies cabling to a wiring closet, while the wiring closets of different buildings are themselves interconnected using a partial mesh or a bus-like backbone to provide redundancy and high capacity between buildings. This hybrid approach is what allows large organisations to obtain the easy fault isolation of a star topology at the department level together with the improved backbone resilience associated with mesh-like interconnection at the campus level. The choice of topology therefore should never be made in isolation but always in conjunction with the expected traffic pattern, the acceptable downtime for the organisation, the budget available for cabling and active networking equipment, and the anticipated rate of future expansion of the network. An examiner setting a question on this topic often expects the student to justify the final recommendation with reference to these practical constraints rather than to simply list the four basic topologies mechanically without properly connecting the chosen topology to a realistic organisational deployment scenario faced by the network designer in practice.